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Final Report

Fairness and Explainability in Machine Learning Models for Young Driver Car Insurance Risk Prediction

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COMP3931 Individual Project

The candidate confirms that the following have been submitted.

Items	**Format**	**Recipient(s) and Date**
Final Report	PDF file	Uploaded to Minerva (29/04/26)
Participant consent forms & Questionnaire responses	ZIP file	Uploaded to Minerva (29/04/26)
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Link to deployed application	URL – `https://sc23ss2-fyp-insurance-fairness.hf.space/`	Sent to supervisor and assessor (29/04/26)
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Summary

Motor insurance premiums for young drivers in the UK are among the highest in Europe, with age frequently cited as the primary justification for elevated pricing. However, the fairness and transparency of this practice have rarely been examined using modern machine learning methods. This project addresses this gap by developing a transparent and fair machine learning framework for young driver motor insurance risk prediction, using publicly available UK road safety data.

The framework is constructed using the 2024 Department for Transport STATS19 dataset, resulting in a cleaned and validated dataset comprising 13,376 observations of young drivers aged 17 to 25. Three supervised classification models were trained and evaluated: Logistic Regression, Random Forest, and XGBoost. Ensemble models achieve a ROC-AUC of 0.689, substantially outperforming the linear baseline of 0.629. A six-stage pipeline implementing iterative leakage detection, threshold optimisation, and stratified evaluation ensured the validity and reproducibility of all results.

SHAP explainability analysis revealed that `age_of_driver` ranks only 13th out of 49 features in terms of predictive importance. Road environment features, including police force area, number of vehicles involved, vehicle type, and speed limit, dominate the predictions. This finding directly challenges the justification for age-based insurance pricing. The model indicates that the location and manner of driving are more significant predictors of risk than the driver's age.

Fairness evaluation using Fairlearn's MetricFrame exposed a consistent performance–fairness tradeoff across all three models. No single model achieved fairness across both sex and age band dimensions simultaneously. Intersectional analysis reveals that disparities compound when both protected attributes are considered together, a pattern that is not visible when attributes are audited independently. Bootstrap resampling confirmed that all fairness gaps are statistically stable.

The findings were made accessible through an interactive Streamlit application, deployed publicly via Hugging Face Spaces. This application provides real-time risk predictions, SHAP waterfall explanations, and a traffic light fairness audit. An informal usability evaluation involving three young drivers confirmed that the application successfully communicates model behaviour to non-technical users. The interface and fairness indicators received maximum usability scores.

All six project objectives were achieved. The framework demonstrates that transparent and fair ML can be applied to insurance risk prediction using publicly available data, and that this approach reveals important questions regarding whether age-based pricing practices are justified by the evidence.

Acknowledgements

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Most of all, I owe everything to my parents. My mother has always been my biggest supporter since my first day at university, always believing in me, guiding me through every challenge, and standing by me as I started my journey into the professional world. My father's steady encouragement and strength have given me the confidence to get to where I am today. I could not have reached this point without them.

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Chapter 1

Introduction and Background Research

1.1 Introduction

Motor insurance plays a critical role in enabling mobility and financial protection for drivers in the UK. However, insurance premiums for young drivers remain disproportionately high compared to other demographic groups. Drivers aged between 17 and 25 consistently face some of the highest motor insurance costs in the UK, largely due to their statistically higher accident rates and claim frequencies [1]. Recent industry data also shows a sharp increase in insurance costs for young drivers in recent years, as illustrated in Figure 1.1, where premiums for 18–24 year olds rise significantly compared to older age groups [2]. For many young people, owning a car is essential rather than optional. High insurance costs can therefore limit their ability to travel to work, attend education or training, and live independently, making motor insurance a significant social and economic issue.

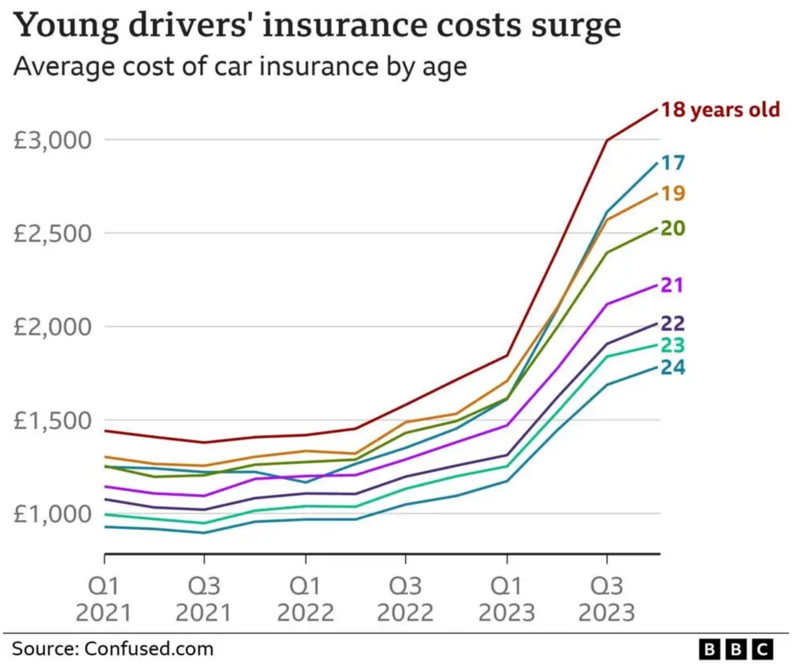

Figure 1.1: Average cost of car insurance by age in the UK between 2021 and 2023, showing significantly higher premiums for young drivers. Source: Confused.com, reproduced by BBC News [2].

This challenge is not only theoretical but is widely experienced in practice. Many young drivers struggle to understand why they are charged substantially higher premiums despite having limited driving history or safe driving behaviour. This lack of transparency can lead to frustration and a perception of unfair treatment, highlighting the need for insurance systems that are not only accurate but also understandable and justifiable to customers.

In recent years, machine learning has transformed insurance pricing, raising three important questions examined in the literature review that follows: how transparent models are in explaining their decisions (Section 1.2.3), whether they treat different groups fairly (Section 1.2.2), and how these concerns apply specifically to young driver insurance (Section 1.2.1).

This project investigates how fairness and explainability can be integrated into machine learning models for young driver motor insurance risk prediction. By developing a reproducible pipeline, training multiple supervised learning models, and evaluating them using both performance and fairness criteria, the project demonstrates how transparent and responsible machine learning can be applied in an insurance context. The work is intended as a proof-of-concept academic study that highlights the trade-offs between accuracy, fairness, and interpretability, rather than a deployable commercial pricing system.

1.2 Literature review

This section reviews existing research relevant to motor insurance risk prediction, machine learning approaches in insurance, fairness in algorithmic decision-making, and explainable artificial intelligence. The aim is to position this project within the current academic and industry landscape, identify limitations of existing approaches, and explain how this project addresses those gaps.

1.2.1 Machine learning in motor insurance risk prediction

Machine learning has become increasingly influential in the insurance industry, particularly for pricing, underwriting, and claims prediction. Traditional actuarial models, such as generalised linear models (GLMs), have long been used to estimate insurance risk due to their interpretability and strong statistical foundations. However, these models often struggle to capture non-linear relationships and complex interactions between variables [3].

To address these limitations, researchers and insurers have adopted supervised machine learning models including decision trees, random forests, gradient boosting methods, and neural networks. In particular, ensemble models such as Random Forest and XGBoost are widely reported to outperform simpler models in predictive tasks due to their ability to capture complex, non-linear interactions in large datasets. Masello et al. [4] demonstrated this in a driver risk assessment framework that integrated XGBoost with SHAP explanations to compute individual weekly insurance premiums using telematics and Advanced Driver Assistance Systems data. While their work confirms the value of combining machine learning with explainability for individual driver risk assessment, two significant limitations remain. (1) the study relies on proprietary telematics data from a commercial fleet operator, which is not publicly available and cannot be independently reproduced. (2) it does not evaluate demographic fairness across protected attributes such as age or sex. Obtaining equivalent telematics data was not feasible for this project due to commercial sensitivity and access restrictions. Instead, this project uses publicly available UK road safety data from the

Department for Transport as a reproducible alternative, and explicitly addresses the fairness gap that Masello et al. leave unexamined.

Despite these performance gains, the increasing use of complex models has raised concerns regarding transparency and accountability in automated pricing decisions. Within the UK market specifically, England et al. [5] modelled motor insurance customer behaviour using agent-based simulation, demonstrating how word-of-mouth networks influence market dynamics, however, their work does not address individual risk prediction, demographic fairness, or model explainability, further highlighting the gaps this project addresses.

1.2.2 Fairness and discrimination in insurance models

Fairness in algorithmic decision-making has become a major area of research, especially in high-stakes domains such as finance, healthcare, and insurance. In the context of motor insurance, unfairness may arise when certain demographic groups are systematically charged higher premiums without sufficient justification based on actual risk. While Barocas and Selbst [6] provide a foundational legal and theoretical analysis of algorithmic discrimination, their work follows the widespread adoption of ML fairness toolkits and does not demonstrate practical measurement of fairness in insurance pricing pipelines.

A key challenge is proxy discrimination, where models rely on variables that are correlated with protected characteristics, even if those characteristics are explicitly excluded. For example, postcode, vehicle type, or employment status may act as proxies for ethnicity or socioeconomic background. Lindholm et al. [7] formalise this problem mathematically, proposing discrimination-free pricing formulas that explicitly remove both direct and indirect discrimination from insurance models. However, their approach requires collecting policyholders' protected characteristics, raising practical privacy concerns. This project takes a complementary approach, applying post-hoc fairness auditing via Fairlearn rather than modifying the pricing formula itself.

To address these issues, researchers have proposed quantitative fairness metrics including demographic parity difference (DPD) and equal opportunity difference (EOD), which measure disparities in model outcomes across groups [8]. However, Hardt et al. demonstrate their framework primarily in a credit scoring context and do not address motor insurance specifically, nor do they consider intersectional fairness across multiple protected attributes simultaneously, a limitation that this project addresses through combined sex and age band analysis using Fairlearn.

1.2.3 Explainable AI in insurance

Explainable Artificial Intelligence (XAI) focuses on making machine learning models more transparent and interpretable to humans. This is particularly important in regulated industries such as insurance, where automated decisions can have significant financial and social consequences for individuals [9].

The distinction between these two approaches is illustrated in Figure 1.2 [10], which contrasts black-box models that produce predictions without explanation against interpretable models that surface the reasoning behind each decision [10]. Among the most widely used XAI techniques is SHapley Additive exPlanations (SHAP), which provides a unified framework for explaining model predictions by attributing contributions to individual input features [11].

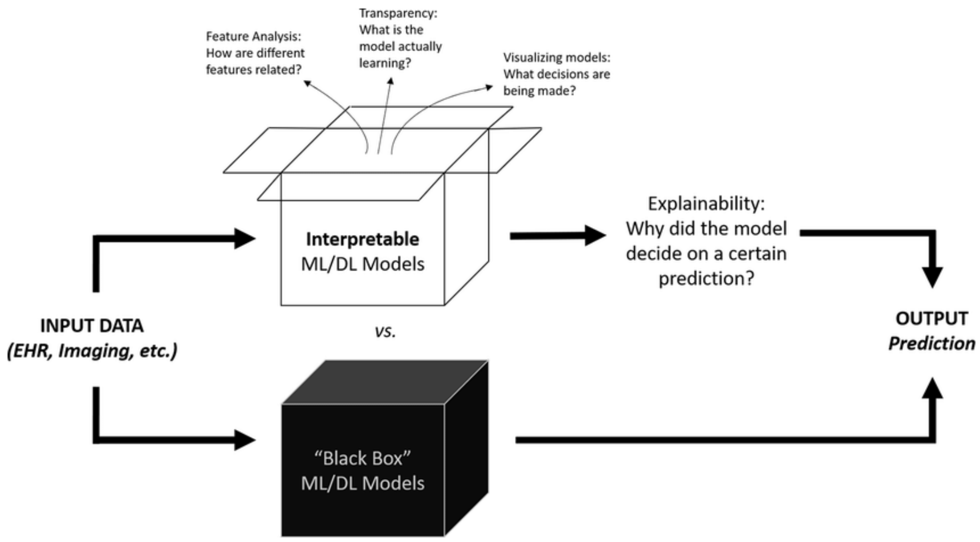

Figure 1.2: Comparison between black-box machine learning models and interpretable/explainable models, highlighting transparency and explainability in decision-making.

SHAP has been applied in insurance risk modelling to interpret both global model behaviour and individual predictions [4]. However, these applications treat explainability and fairness as separate concerns, SHAP reveals *what* the model predicts but does not evaluate *whether* those predictions treat demographic groups equitably. This project addresses that gap by combining SHAP with Fairlearn fairness metrics within a single pipeline, enabling simultaneous assessment of both model transparency and demographic equity

1.2.4 Research gap and motivation

While existing research has made significant contributions to individual aspects of insurance risk modelling, several important gaps remain when considered collectively. First, studies applying machine learning to motor insurance, such as Masello et al. [4] and Poufinas et al. [12], focus primarily on predictive accuracy without evaluating whether model outcomes are fair across demographic groups. Second, fairness-focused work such as Hardt et al. [8] and Lindholm et al. [7] provides theoretical frameworks for non-discrimination but does not demonstrate practical implementation within a reproducible supervised machine learning pipeline. Third, while SHAP has been applied to insurance risk models [4], it is not combined with quantitative fairness auditing in a single framework. Finally, no existing study specifically targets young driver motor insurance risk prediction in the UK using publicly available road safety data.

This project addresses all four gaps simultaneously by: (1) building a reproducible pipeline using real UK road safety data from the Department for Transport; (2) training and comparing three supervised learning models — Logistic Regression, Random Forest, and XGBoost; (3) applying SHAP for individual-level explainability, revealing which features most strongly influence risk predictions; and (4) evaluating fairness using Demographic Parity Difference and Equalized Odds Difference across sex, age band, and their intersection. The combination of these four elements within a single transparent framework, demonstrated through an interactive Streamlit application, represents the primary contribution of this work.

1.3 Project Aim, Objectives and Deliverables

1.3.1 Project Aim

The aim of this project is to design, implement, and evaluate a transparent and fair machine learning framework for predicting motor insurance risk for young UK drivers (aged 17–25) using publicly available road safety data. Predictive performance is evaluated alongside explainability and fairness auditing by integrating supervised machine learning with XAI techniques and quantitative fairness metrics. The work is intended as a proof-of-concept academic study highlighting trade-offs between accuracy, fairness, and interpretability, rather than a deployable commercial pricing system.

1.3.2 Project Objectives

To achieve the stated aim, the project is structured around the following objectives:

- **Objective 1:** To construct a representative dataset of young driver collision records by integrating publicly available UK road safety data from the Department for Transport, ensuring ethical compliance and reproducibility.
- **Objective 2:** To develop a reproducible data-processing pipeline that cleans, encodes, and prepares the dataset for supervised machine learning, ensuring repeatability and robustness of results.
- **Objective 3:** To train and evaluate multiple supervised machine learning models, including Logistic Regression, Random Forest, and XGBoost, for predicting insurance claim risk among young drivers using standard performance metrics.
- **Objective 4:** To analyse model behaviour using explainable artificial intelligence techniques, specifically SHapley Additive exPlanations (SHAP), in order to understand how individual features influence predictions.
- **Objective 5:** To assess algorithmic fairness by computing and interpreting quantitative fairness metrics, such as Demographic Parity Difference and Equal Opportunity Difference, and to identify whether bias arises from the data, the model, or proxy variables.
- **Objective 6:** To develop a simple interactive Streamlit-based demonstration that presents model predictions, explanations, and fairness insights to a non-technical audience.

1.3.3 Project Deliverables

The project will produce two main deliverables:
D1. Project Report – A comprehensive written report documenting the full project lifecycle, including background research, dataset construction, methodology, model development, explainability and fairness analysis, and evaluation of results.

D2. Software Repository – A well-structured GitHub repository containing all technical artefacts produced during the project, including:

- **D2.1. Dataset acquisition and construction pipeline** integrating publicly available UK road safety data with derived insurance-relevant features to form a representative motor insurance risk dataset. (Linked to Objective 1)
- **D2.2. Reproducible data-processing pipeline** for cleaning, encoding, and preparing the dataset for supervised machine learning experiments. (Linked to Objective 2)
- **D2.3. Trained machine learning models and evaluation**, including Logistic Regression, Random Forest, and XGBoost, with standard performance metrics such as Accuracy, F1-score, and ROC-AUC. (Linked to Objective 3)
- **D2.4. Explainability and fairness analysis outputs**, including SHAP-based model interpretations and quantitative fairness metrics such as Demographic Parity Difference and Equal Opportunity Difference. (Linked to Objectives 4 and 5)
- **D2.5. Interactive Streamlit demonstration application** enabling non-technical users to explore predictions, explanations, and fairness insights. (Linked to Objective 6)

1.4 Project Scope and Contributions

This project focuses on how machine learning can assess motor insurance risk for young drivers in a way that is accurate, fair, and understandable, examining *how* risk predictions are produced and *whether* they disadvantage young drivers through hidden bias or proxy variables. Rather than replicating commercial pricing systems, it adopts an experimental data-driven approach using publicly available UK road safety data, ensuring ethical compliance throughout.

The contribution lies not in producing a deployable insurance product, but in demonstrating how algorithmic decisions affecting young drivers can be examined, questioned, and made more transparent. This is of growing importance given the UK Financial Conduct Authority's requirements for fair value in insurance pricing [13] and the EU AI Act's classification of insurance pricing systems as high-risk AI applications requiring transparency and fairness auditing [14]. By demonstrating that such auditing is achievable using only publicly available data, this project contributes a proof-of-concept that is directly relevant to the emerging regulatory landscape.

Chapter 2

Methodology

2.1 Methodological Framework and Project Design

This project is structured around a fairness-aware machine learning methodology for analysing motor insurance risk among young drivers. Rather than treating model development as a single optimisation task, the methodology is designed as a sequence of explicit and auditable stages, each of which influences predictive performance, fairness outcomes, and interpretability.

The core design choice is to adopt an *iterative machine learning pipeline*, inspired by established data mining frameworks such as CRISP-DM, but adapted to reflect modern data science practice and the specific requirements of fairness and explainability analysis. This approach is particularly well-suited to the problem domain, where insights gained during evaluation or bias analysis may require earlier decision, such as feature selection or target construction, to be revisited.

Three core principles guide the methodological design:

1. The workflow must be reproducible and transparent, allowing each transformation from raw data to model output to be inspected, traced, and justified.
2. Fairness and explainability are treated as integral components of the solution design rather than optional checks applied after modelling.
3. The methodology explicitly supports iteration, recognising that responsible machine learning requires refinement based on empirical evidence rather than a rigid, one-pass process.

Figure 2.1 provides a high-level overview of the adopted methodology, illustrating the main stages of the pipeline and the iterative feedback loops between data preparation, modelling, evaluation, and fairness analysis. Each stage shown in the diagram is discussed in detail in the following subsections.

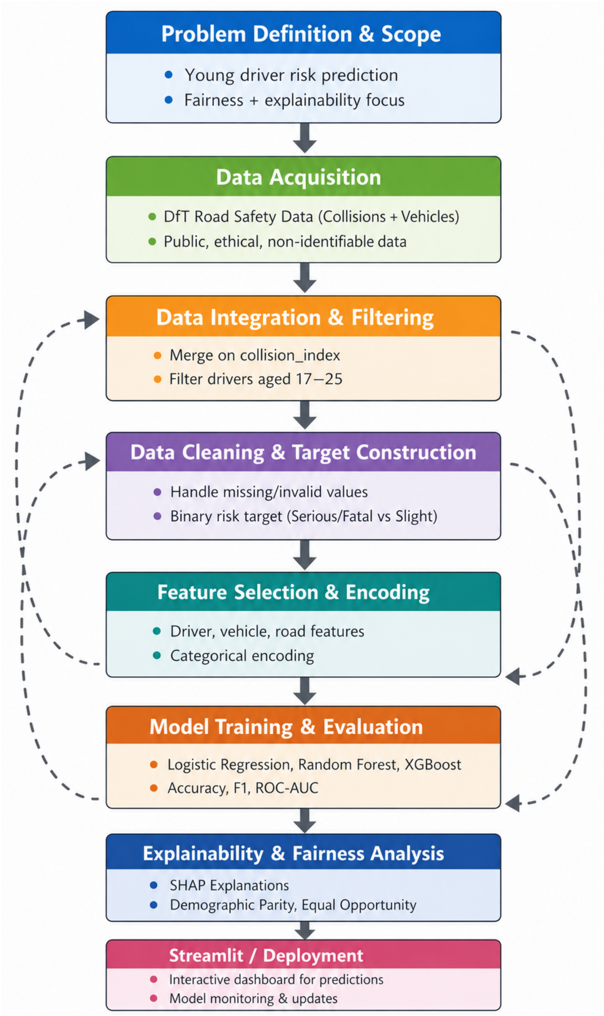

Figure 2.1: Overview of the fairness-aware machine learning pipeline adopted in this project. Solid arrows indicate the primary workflow, while dashed arrows represent iterative refinement based on evaluation, explainability, and fairness analysis.

2.1.1 Problem Definition and Scope

This project predicts motor insurance risk for young UK drivers aged 17–25[1], a demographic consistently facing disproportionately high premiums. The problem is framed as a supervised binary classification task based on observable road collision outcomes, enabling controlled investigation of predictive behaviour, fairness, and explainability without replicating commercial pricing systems.

The scope is constrained in three ways: (1) the analysis is restricted to drivers aged 17–25; (2) the target distinguishes higher-severity collisions (fatal or serious) from lower-severity ones (slight); and (3) live pricing deployment is excluded in favour of analytical insight and responsible model evaluation.

Protected attributes — sex and age band of driver — are identified at this stage and deliberately excluded from model inputs, ensuring systematic fairness assessment throughout

[1] The Streamlit demonstration uses DfT STATS19 age band coding where band 4 covers drivers aged 16–20. The model was trained exclusively on drivers aged 17–25, as 16-year-olds cannot hold a full UK driving licence. Any age 16 input in the application uses modal default values and is illustrative only.

the pipeline. This framing ensures the project evaluates not only how accurately risk can be predicted, but whether prediction outcomes differ across demographic groups and why.

2.1.2 Data Acquisition

During the early stages of the project, publicly available UK motor insurance datasets were explored as a potential data source. However, no suitable dataset was found that allowed independent analysis of risk, fairness, and explainability without access restrictions or proprietary constraints. The project therefore adopts a data construction approach, inferring insurance-style risk from observed road traffic collision outcomes using the Department for Transport (DfT) road safety dataset, released under the Open Government Licence [15].

Two datasets from the 2024 release are used [16]:

Collision-level dataset Contextual information about each recorded accident including severity, road conditions, location, date, and environmental factors.

Vehicle-level dataset Driver characteristics and vehicle attributes including age, sex, vehicle type, and manoeuvre.

This source was selected because the data is collected under a nationally standardised specification (STATS19) ensuring internal consistency; the vehicle-level structure mirrors the tabular format used in insurance risk modelling; and the data is fully anonymised and publicly available, ensuring ethical compliance. The analysis focuses on a single validated year (2024) to avoid complications from changes in data definitions over time. This directly supports Objective 1.

2.1.3 Data Integration and Understanding

After data acquisition, the collision-level and vehicle-level datasets needed to be reliably linked. Table 2.1 summarises the evaluation of candidate join keys.

Table 2.1: Evaluation of candidate join keys for dataset integration

Candidate Key	**Issue Identified**	**Decision**
collision_year	Not unique; repeated across records	Rejected
collision_ref_no	Reused across police forces	Rejected
collision_index	Unique per collision; shared across datasets	Selected

The datasets were therefore merged using an inner join on `collision_index`, confirming the one-to-many relationship illustrated in Figure 2.2. A single collision may involve multiple vehicles, making this structure well-suited to insurance-style risk modelling where risk is assessed at the individual driver level. Validation checks confirmed expected row counts, relationship consistency, and the absence of unintended duplication. This directly supports Objective 1.

2.1.4 Filtering to Young Drivers

Following dataset integration, the analysis was restricted to drivers aged between 17 and 25 inclusive, retaining only rows satisfying $\texttt{17} \leq \texttt{age\_of\_driver} \leq \texttt{25}$. This filtering step aligns

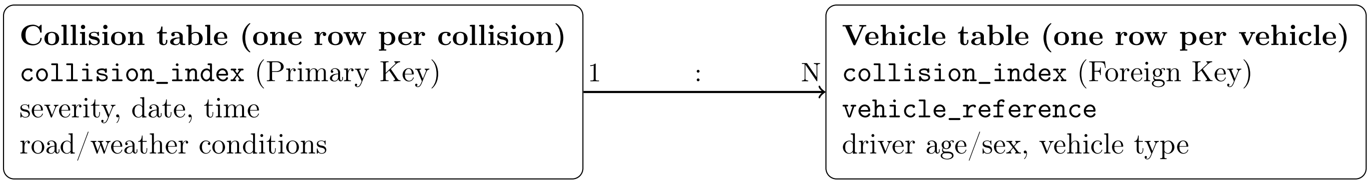

Figure 2.2: One-to-many relationship between collision-level and vehicle-level datasets using `collision_index`.

directly with the project scope and ensures that all modelling, explainability, and fairness assessments are applied to a clearly defined and policy-relevant demographic group. Modelling the full population would introduce unnecessary heterogeneity and dilute the fairness analysis. This produced 13,709 observations prior to fairness-specific cleaning, with summary statistics confirming the minimum age was 17 and maximum was 25.

2.1.5 Data Cleaning and Target Construction

After demographic filtering, the dataset was subjected to systematic cleaning to ensure that modelling decisions were based on valid and interpretable records.

Handling Invalid and Non-Informative Codes

The DfT road safety dataset uses numeric coding schemes to represent categorical values. Certain codes indicate missing, unknown, or out-of-range values (e.g., `-1` or `3 = Not known` in `sex_of_driver`).
For fairness analysis to be meaningful, protected attributes must represent clearly defined and interpretable groups. Therefore:

- `sex_of_driver = 1 (Male)` and `2 (Female)` were retained.
- `3 (Not known)` and `-1 (Missing / out of range)` were removed.

This produced a final fairness-aware dataset of **13,376** observations for all subsequent modelling, explainability, and fairness analysis. Removing ambiguous protected attribute values maintains the integrity of fairness metrics and ensures valid subgroup comparisons.

Construction of the Risk Target Variable

The original `collision_severity` variable (1 = Fatal, 2 = Serious, 3 = Slight) was transformed into a binary risk indicator, as illustrated in Figure 2.3.

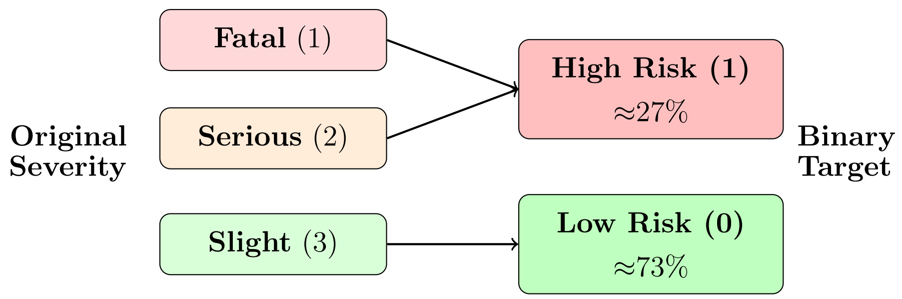

Figure 2.3: Construction of the binary risk target variable from the original DfT collision severity encoding.

This design choice was made for three reasons: (1) fatal collisions are extremely rare within the young-driver subset making multi-class modelling unstable; (2) insurance risk assessment typically distinguishes between higher and lower severity outcomes; and (3) a binary formulation simplifies interpretation of fairness metrics such as Demographic Parity and Equal Opportunity Difference. The resulting moderate class imbalance (73% low risk, 27% high risk) reflects real-world collision patterns while remaining suitable for supervised classification. This stage directly supports Objective 2, ensuring all downstream modelling and fairness analysis are grounded in clearly defined and reproducible transformations.

2.1.6 Feature Selection and Encoding

Following data cleaning and target construction, the next stage involved preparing the feature set for supervised modelling. This required careful consideration of which variables should be retained, transformed, or excluded to ensure predictive validity while avoiding data leakage.

Feature Selection Principles

Feature selection was guided by three key principles:

1. **Avoidance of Data Leakage:** Any variable directly encoding or derived from `collision_severity` was excluded to prevent artificial inflation of model performance.
2. **Removal of Identifier Variables:** Unique identifiers (e.g., `collision_index`, `vehicle_reference`) were removed as they carry no predictive signal.
3. **Control of High-Cardinality Variables:** Categorical variables with excessive unique categories (e.g., `generic_make_model`) were excluded to prevent dimensional explosion and maintain interpretability.

After applying these criteria, the final baseline feature matrix consisted of 49 predictors.

Temporal Feature Engineering

The original `time` and `date` variables were stored as strings. To extract meaningful temporal structure:

- `time` was converted to an integer hour variable (0–23), capturing daily traffic patterns.

- `date` was parsed to extract the month (1–12), allowing the model to learn potential seasonal effects.

The original string-based time and date columns were subsequently removed to ensure numerical consistency.

Protected Attributes for Fairness Evaluation

Two protected attributes were separated from the predictive feature matrix and retained exclusively for post-hoc fairness analysis:

`sex_of_driver` `age_band_of_driver`

This prevents the model from explicitly learning on protected attributes while still enabling subgroup evaluation, directly supporting Objective 5.

Final Modelling Structure

The final modelling dataset is summarised below:

13,376 observations | 49 predictive features | 1 binary target: `high_risk`

All remaining predictors were numeric, enabling direct use in linear and tree-based classification models, directly supporting Objective 2.

2.1.7 Model Training and Evaluation

The modelling strategy was designed around three guiding principles: (1) **interpretability** — beginning with linear baselines to establish explainable reference behaviour; (2) **minority-class sensitivity** — prioritising detection of high-risk drivers given the safety implications of false negatives; and (3) **policy-aligned evaluation** — analysing threshold-dependent behaviour rather than relying on default classification rules.

Data Splitting Strategy

The dataset was partitioned using an 80/20 stratified train-test split, preserving the class distribution of `high_risk` to prevent artificial inflation of minority-class performance and ensure realistic evaluation conditions.

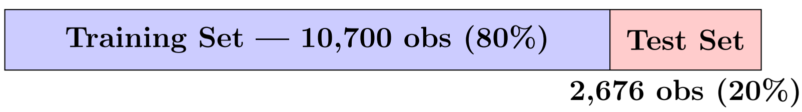

Figure 2.4: Stratified 80/20 train-test split preserving class distribution (73% low risk, 27% high risk).

All evaluation metrics were computed exclusively on the held-out test set to prevent information leakage, directly supporting Objective 3.

Evaluation Metrics

Given the class imbalance (73% low risk, 27% high risk), accuracy alone is misleading — a naive model predicting all drivers as low risk would achieve 73% accuracy without learning any meaningful patterns. Five complementary metrics were therefore used to evaluate model performance:

Accuracy Overall proportion of correct predictions. Reported for completeness but not used as the primary decision metric due to class imbalance.

Precision Of all drivers flagged as high risk, the proportion that are genuinely high risk. A low precision indicates excessive false alarms.

Recall Of all truly high risk drivers, the proportion correctly identified. Prioritised in this project as missing a genuinely dangerous driver carries greater societal cost than incorrectly flagging a safe one.

F1-score Harmonic mean of precision and recall, providing a single balanced measure of minority-class performance. Used as the primary model selection criterion alongside ROC-AUC.

ROC-AUC Area under the Receiver Operating Characteristic curve, measuring the model's ability to rank high-risk drivers above low-risk ones independently of any fixed threshold. Particularly important for comparing models before threshold tuning.

Model comparison was conducted holistically across all five metrics rather than optimising a single measure, ensuring that model selection aligned with both predictive performance and the safety-oriented objectives of this project.

Model Selection

Three model families were evaluated, proceeding from interpretable linear baselines to more complex non-linear ensembles:

Logistic Regression Selected as the linear baseline due to its interpretability and well-established use in binary classification [3]. Three configurations were evaluated: standard, feature-standardised, and class-weight balanced. The balanced configuration substantially increased high-risk recall to approximately 64%, illustrating the trade-off between global accuracy and minority-class sensitivity.

Random Forest Builds an ensemble of decision trees independently, aggregating predictions through majority voting to capture complex feature interactions. Unlike Logistic Regression, `class_weight=balanced` reduced recall slightly, confirming that the right imbalance strategy depends on the model family.

XGBoost Builds trees sequentially, each correcting residual errors of the previous ensemble. Built-in regularisation (`reg_lambda=1.0`) and subsampling (`subsample=0.8`) handle class imbalance without requiring class weighting. Both ensemble models achieved ROC-AUC of 0.689, confirming stronger discrimination than Logistic Regression (0.629).

Why Tree-Based Models Over Neural Networks?

Tree-based ensemble methods were selected over neural network approaches for three reasons, summarised in Figure 2.5.

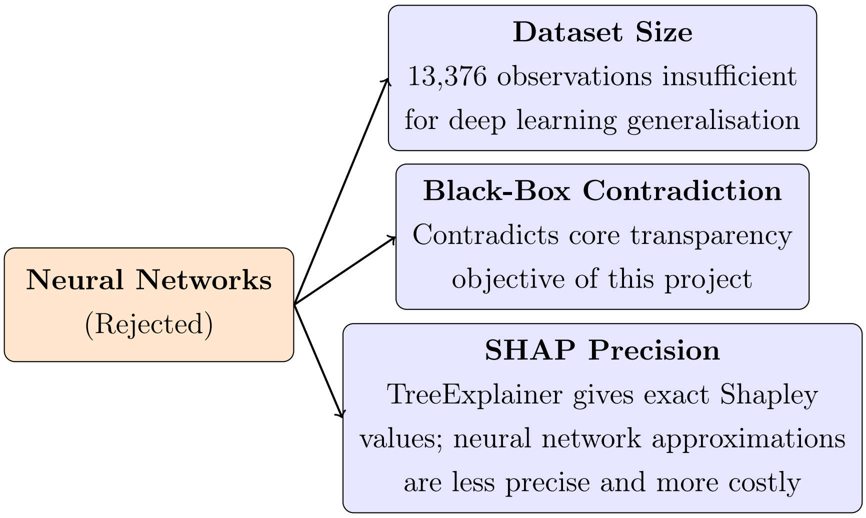

Figure 2.5: Justification for selecting tree-based models over neural network approaches.

This decision directly supports the project's core transparency objective — SHAP's TreeExplainer provides exact Shapley value computation for tree-based models, whereas approximations for neural networks are less precise and more computationally expensive [11].

Threshold Adjustment Strategy

All three models produce class probabilities rather than hard labels. By default, a threshold of $\tau = 0.5$ is applied, meaning a driver is flagged as high risk only if the model assigns a probability of 50% or greater. However, in insurance risk prediction, missing a genuinely high-risk driver carries greater cost than incorrectly flagging a low-risk one — comparable to a fire alarm that should occasionally trigger falsely rather than fail to trigger at all.
For a predicted probability $\hat{p}$, the classification rule is:

$$\hat{y} = \begin{cases} 1 & \text{if } \hat{p} \geq \tau \\ 0 & \text{otherwise} \end{cases}$$

Threshold optimisation was therefore treated as a policy parameter rather than a model hyperparameter, keeping the underlying model unchanged while adjusting the decision boundary to prioritise recall for high-risk drivers. Four thresholds were evaluated ($\tau \in \{0.5, 0.4, 0.3, 0.2\}$) for each model, with detailed results presented in Table C.1 in Appendix C.

2.1.8 Explainability and Fairness Analysis

The final stage of the pipeline applies explainability and fairness analysis to the trained models, reflecting the core objectives of this project.

SHAP Explainability

SHAP (SHapley Additive exPlanations) was applied to the best-performing model to explain both global and individual-level predictions [11]. TreeExplainer was selected over other SHAP variants as it provides exact Shapley value computation for tree-based models, making it both faster and more precise than kernel-based approximations. Two visualisations were generated: a bar chart showing mean absolute SHAP values across all test drivers (global importance), and a beeswarm plot showing individual driver-level feature contributions. This directly addresses Objective 4.

Fairness Evaluation

Fairness was evaluated using Fairlearn's `MetricFrame`, computing two quantitative metrics across protected subgroups:

Demographic Parity Difference (DPD) Measures the difference in positive prediction rates across groups — a high DPD indicates one group is flagged as high risk at a much higher rate than another, regardless of actual risk.

Equal Opportunity Difference (EOD) Measures the difference in true positive rates across groups — a high EOD indicates the model identifies genuinely high-risk drivers more effectively for one group than another.

Both metrics were computed across three dimensions: sex of driver, age band, and their intersection (sex $\times$ age band). Bootstrap resampling (200 iterations) was applied to confirm that observed fairness gaps were statistically stable rather than artefacts of the test set. This directly addresses Objective 5.

2.1.9 Streamlit Demonstration

The insights generated from both explainability and fairness analysis are surfaced to non-technical users through an interactive Streamlit demonstration application, deployed publicly via Hugging Face Spaces. The application accepts road and vehicle condition inputs, generates a risk prediction, produces a driver-specific SHAP waterfall explanation, and displays pre-computed fairness metrics using a traffic light indicator system. This directly addresses Objective 6.

2.2 Project Management and Engineering Practices

Given the iterative nature of fairness-aware machine learning, structured project management was essential. An Agile approach was adopted using a GitHub Kanban board, decomposing the project into incremental sprints aligned with each pipeline stage. This allowed new tasks, such as additional leakage detection rounds, to be rapidly incorporated when empirical findings demanded iteration, reflecting the CRISP-DM-inspired methodology described in Section 2.1.

2.2.1 Repository Structure and Version Control

All code was organised into modular Jupyter notebooks, each corresponding to a distinct pipeline stage. Table 2.2 summarises the repository structure and the purpose of each component.

Table 2.2: Modular notebook structure of the project repository

Notebook	**Purpose**
`01_data_loading_merge_audit`	Data ingestion, relational validation, and dataset integration
`02_feature_engineering`	Cleaning, leakage prevention, and dataset preparation
`03_baseline_model`	Logistic Regression baselines and class imbalance experiments
`04_tree_baseline_models`	Random Forest and XGBoost implementation
`05_fairness_metrics`	Demographic Parity and Equal Opportunity analysis
`06_shap_explanations`	Post-hoc explainability analysis

The full repository structure, including the `src/`, `tests/`, `models/`, and `data/` directories, is shown in Figure C.7 in Appendix C.
Development was conducted using Git with a professional branching strategy — each experiment was isolated on a dedicated feature branch and merged into the main branch only after validation. Fixed random seeds (`random_state=42`) were applied consistently across all experiments to guarantee reproducibility.

2.2.2 Agile Planning via Kanban

Project tasks were managed using a GitHub Kanban board decomposed into five progressive stages, as illustrated in Table 2.3.

Table 2.3: GitHub Kanban workflow stages used throughout the project

Stage	**Description**
Backlog	All identified tasks awaiting prioritisation
Ready	Tasks fully scoped and ready to begin
In Progress	Tasks actively under development
In Review	Completed tasks undergoing validation and testing
Done	Validated and merged tasks

Each modelling component was decomposed into specific actionable tasks encouraging incremental development. When evaluation revealed issues such as iterative leakage detection (Chapter 3), new tasks were immediately added and prioritised. The final Kanban board, showing all tasks successfully completed in the Done column, is shown in Figure C.8 in Appendix C.

Chapter 3

Implementation and Validation

3.1 System Overview

This chapter describes the practical implementation of the fairness-aware machine learning framework introduced in Chapter 2, covering specific technical choices, library configurations, and validation outcomes.
The framework was implemented entirely in Python 3.12.7, with all development conducted within a modular Jupyter notebook environment and version-controlled using Git. The full pipeline spans six sequential stages, illustrated in Figure 2.1

The core technology stack is summarised in Table 3.1. Library versions are recorded to ensure reproducibility.

Table 3.1: Technology stack used in the implementation.

Library	**Version**	**Role**
Python	3.12.7	Core programming language
pandas	2.3.3	Data loading, cleaning, and transformation
scikit-learn	1.8.0	Logistic Regression, Random Forest, evaluation
XGBoost	3.1.3	Gradient boosted classification
SHAP	0.50	Post-hoc feature importance and explainability
Fairlearn	0.13.0	Fairness metrics and subgroup evaluation
Streamlit	1.53.1	Interactive web application
Git	–	Version control and reproducibility

While the pipeline was successfully implemented end-to-end, three significant challenges were encountered during development, each requiring iterative investigation and resolution, as described below.

Implementation Challenges and Resolutions

1. **Iterative Data Leakage Detection:** Initial configurations achieved ROC-AUC exceeding 0.95, indicating post-accident features had leaked into the feature matrix. Three iterative rounds of leakage analysis and model retraining were required to resolve this, reducing ROC-AUC to a realistic 0.689. Full validation is presented in Table 3.10.

2. **Class Imbalance Asymmetry:** Standard imbalance techniques produced inconsistent results across model families. `class_weight=balanced` improved Logistic Regression recall from 3% to 59% but reduced Random Forest recall slightly. This was resolved by applying model-specific strategies: class weighting for Logistic Regression and threshold tuning for tree-based models, as documented in Section 3.3.

3. **SHAP Dimensionality:** SHAP's `TreeExplainer` produced three-dimensional output arrays for tree-based models. This was resolved by implementing an explicit shape check extracting the high-risk class dimension (`shap_values[:, :, 1]`), validated in `test_shap.py`.

3.2 Data Processing Pipeline Implementation

The preprocessing pipeline was implemented in `02_feature_engineering_and_preprocessing.ipynb`, taking as input the fairness-cleaned dataset `df_young_fair_2024.csv` (13,376 observations, 76 columns) produced during data integration. The pipeline consisted of four sequential operations: column removal, temporal feature engineering, high-cardinality variable exclusion, and final dataset construction.

Column Removal and Leakage Prevention

The first operation removed variables that were unsuitable for modelling. These fell into five categories, summarised in Table 3.2.

Table 3.2: Variables removed during preprocessing and the rationale for each.

Variable(s)	Category	Rationale
`collision_index`, `collision_ref_no`, `vehicle_reference`	Identifier	No predictive signal; risk of memorisation
`collision_severity`, `enhanced_severity_collision`, `collision_adjusted_severity_serious`, `collision_adjusted_severity_slight`	Data leakage	Directly encode the target variable
`location_easting_osgr`, `location_northing_osgr`, `longitude`, `latitude`	Raw geolocation	High cardinality; not meaningful without spatial aggregation
`lsoa_of_driver`, `lsoa_of_accident_location`	Area identifier	High cardinality area codes unsuitable for baseline modelling
`vehicle_location_restricted_lane_historic`	Duplicate/Historic	Redundant historic version of an existing variable

The severity columns directly describe the outcome the model is trying to predict (`high_risk`), keeping them would be like giving a student the exam answers in advance. However, after the first model iteration, results looked suspiciously good, suggesting further leakage variables remained. This prompted a second round of iteration, documented in the following subsection.

Iterative Leakage Detection

The second iteration revealed three additional leakage variables and two uninformative variables that had not been identified in the initial column selection, summarised in Table 3.3.

Table 3.3: Additional variables removed following iterative leakage analysis.

Variable	Category	Rationale
number_of_casualties	Data leakage	Post-accident outcome; not observable at prediction time
did_police_officer_ attend_scene_of_accident	Data leakage	Severity proxy; correlates strongly with fatal/serious outcomes
collision_injury_based	Data leakage	Injury-based encoding derived directly from the target variable
collision_year_y	Duplicate	Redundant year column from merging
escooter_flag	Near-zero variance	Overwhelmingly constant; no predictive signal

After each removal, the model was retrained and ROC-AUC was monitored. Each drop in performance confirmed that earlier configurations had been reading post-accident information rather than learning genuine road safety patterns. The full validation of this iterative process is presented in Table 3.10 in Section 3.7.

Temporal Feature Engineering

The dataset contained two string-formatted temporal variables, `time` and `date`, which could not be used directly as model inputs. These were transformed as follows:

- `time` was parsed using `pd.to_datetime` with format `'%H:%M'` and the integer hour (0–23) was extracted, capturing diurnal traffic patterns relevant to collision risk.
- `date` was parsed using `pd.to_datetime` with `dayfirst=True` and the integer month (1–12) was extracted, allowing the model to capture seasonal variation in driving conditions.

High-Cardinality Variable Exclusion

Four categorical variables contained too many unique categories to be used safely in baseline modelling, one-hot encoding them would have introduced hundreds of binary columns, causing dimensional instability and reducing interpretability:

`generic_make_model` `local_authority_ons_district`

`local_authority_highway` `local_authority_highway_current`

These were excluded from the baseline feature matrix, with their potential inclusion noted as a direction for future work.

Final Dataset Construction

Following all preprocessing operations, the cleaned dataset was split into three distinct objects and partitioned for modelling, as illustrated in Figure 3.1.

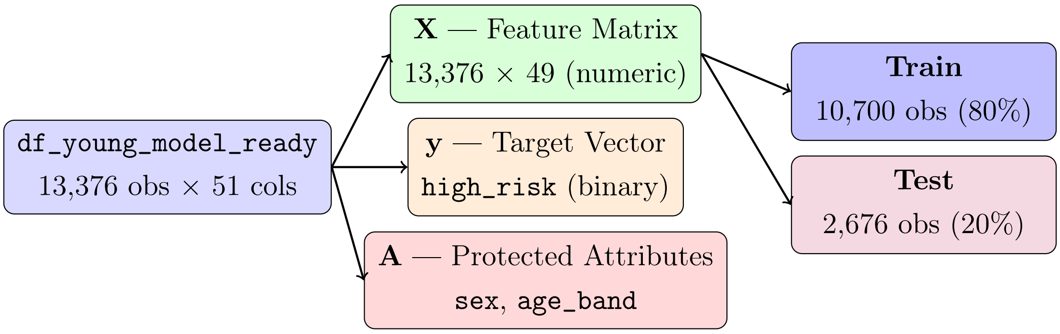

Figure 3.1: Final dataset construction showing the three modelling objects and stratified 80/20 train-test split (`test_size=0.2`, `stratify=y`, `random_state=42`).

A was kept separate from **X** to prevent the model learning directly on protected attributes, while preserving them for post-hoc fairness auditing. Stratification preserved the 73%/27% class distribution in both partitions, preventing artificial skew in evaluation metrics.

3.3 Model Implementation

Three classification models were implemented using scikit-learn and XGBoost, each configured to address the class imbalance (~73% low risk, ~27% high risk) inherent in the dataset. All models were trained on the same stratified train-test split (10,700 training, 2,676 test observations, `random_state=42`) to ensure comparability.

Logistic Regression

Three Logistic Regression configurations were evaluated using scikit-learn's `LogisticRegression`:

1. **Unscaled** — `max_iter=1000`, no preprocessing. Produced a convergence warning, indicating that feature magnitudes were affecting optimisation stability.
2. **Scaled** — Features standardised using `StandardScaler` within a scikit-learn `Pipeline`, `max_iter=5000`. Resolved the convergence warning and marginally improved performance.
3. **Balanced** — `StandardScaler` combined with `class_weight='balanced'` and `max_iter=5000`. This configuration re-weights the loss function inversely proportional to class frequencies, penalising misclassification of high-risk drivers more heavily. It was selected as the final Logistic Regression configuration, producing the best minority-class recall (0.59) and ROC-AUC (0.629).

Using a scikit-learn `Pipeline` ensured that `StandardScaler` was fitted exclusively on training data, preventing test-set information from influencing feature normalisation.

Random Forest and XGBoost

Both models were implemented using their respective scikit-learn and XGBoost classes. Table 3.4 summarises the hyperparameters used for each.

Table 3.4: Hyperparameters for Random Forest and XGBoost.

Parameter	Random Forest	XGBoost
`n_estimators`	300	300
`max_depth`	–	4
`learning_rate`	–	0.05
`subsample`	–	0.8
`colsample_bytree`	–	0.8
`reg_lambda`	–	1.0
`eval_metric`	–	`logloss`
`class_weight`	`balanced`	–
`n_jobs`	-1	–
`random_state`	42	42

For Random Forest, both balanced and unbalanced configurations were evaluated; the balanced configuration was selected as it produced more informative minority-class behaviour, though ROC-AUC remained identical (0.689). For XGBoost, a shallow tree depth (`max_depth=4`) limited overfitting, while row and feature subsampling (`subsample=0.8`, `colsample_bytree=0.8`) and L2 regularisation (`reg_lambda=1.0`) further controlled model complexity.

Threshold Optimisation

All three models produce class probabilities rather than hard labels. The default threshold of $\tau = 0.5$ was evaluated alongside $\tau \in \{0.4, 0.3, 0.2\}$. At $\tau = 0.5$, both ensemble models were heavily biased towards the majority class, correctly identifying fewer than 14% of high-risk drivers. Lowering the threshold to $\tau = 0.3$ substantially improved this — recall rose to 0.614 for Random Forest and 0.518 for XGBoost — while keeping precision above 0.39. This reflects the core insurance insight that missing a genuinely high-risk driver is costlier than occasionally flagging a safe one. Full threshold evaluation results are presented in Table C.1 in Appendix C.

Model Comparison

The final configurations of all three models at their selected thresholds are compared in Table 3.5.

Table 3.5: Comparison of final model configurations at selected decision thresholds (metrics reported for the high-risk class).

Model	Threshold	Accuracy	Recall	Precision	F1	ROC-AUC
Logistic Regression	0.5	0.578	0.59	0.33	0.427	0.629
Random Forest	0.3	0.650	0.614	0.397	0.482	0.689
XGBoost	0.3	0.691	0.518	0.431	0.471	0.689

Both ensemble models substantially outperformed Logistic Regression in ROC-AUC (0.689 vs 0.629), reflecting their capacity to capture non-linear feature interactions. XGBoost achieved the highest accuracy (0.691) and precision (0.431), while Random Forest produced the highest recall (0.614), identifying a greater proportion of true high-risk drivers at the cost of more false positives. The equal ROC-AUC of both ensemble models suggests comparable discriminative ability at the probability level, with the performance difference emerging only after threshold application. Confusion matrices are presented in Appendix C (Figures C.1, C.2, and C.3) and ROC curves in Chapter 4 (Figure 4.1).

3.4 SHAP Explainability Implementation

SHAP explainability was implemented in `06_shap_explanations.ipynb` using `shap` v0.50, applied to XGBoost as the best performing model (accuracy = 0.691, ROC-AUC = 0.689).

TreeExplainer Configuration

SHAP's `TreeExplainer` was selected over kernel-based alternatives for three reasons: (1) it provides exact Shapley value computation rather than approximations; (2) it is significantly faster than `KernelExplainer` on large feature sets; and (3) it is natively optimised for XGBoost's gradient boosted tree structure [11]. The explainer was instantiated and applied to the held-out test set as follows:

```
explainer = shap.TreeExplainer(xgb_model)
shap_values = explainer.shap_values(X_test)
if len(shap_values.shape) == 3:
    shap_values = shap_values[:, :, 1]
```

The shape check extracts the high-risk class dimension from the three-dimensional output array, as described in the implementation challenges in Section 3.1.

Visualisations and Key Finding

Two SHAP visualisations were generated:

Bar chart (global importance) Shows mean absolute SHAP values per feature across all 2,676 test drivers. `police_force` was the strongest predictor (mean |SHAP| = 0.195), acting as a regional proxy for road conditions and accident reporting patterns.

Beeswarm plot (individual effects) Shows one dot per driver per feature, coloured red for high feature values and blue for low. Points spread right indicate increased risk. For example, high `number_of_vehicles` values spread strongly right, confirming multi-vehicle collisions carry significantly elevated risk.

Critically, `age_of_driver` ranked 13th out of 49 features (mean |SHAP| = 0.032), confirming that the model does not primarily use age to predict collision risk. Road environment features dominate the predictions, directly challenging the justification for age-based insurance pricing and representing a core contribution of this project.

3.5 Fairness Evaluation Implementation

Fairness evaluation was implemented in `05_fairness_metrics.ipynb` using the Fairlearn library. The same three models trained in the baseline notebook were retrained identically here, with thresholds applied as a post-processing step: $\tau = 0.5$ for Logistic Regression and $\tau = 0.3$ for both Random Forest and XGBoost, consistent with the threshold selection rationale in Section 3.3.

MetricFrame Setup

Fairlearn's `MetricFrame` computed accuracy, precision, recall, F1, and selection rate per demographic group, alongside DPD and EOD as defined in Section 2.1.8. A value of zero indicates perfect fairness; a high EOD means the model identifies high-risk drivers more effectively for one group than another.

Fairness by Sex

Fairness was first evaluated with `sex_of_driver` as the sensitive attribute (1 = male, 2 = female). Table 3.6 summarises the fairness gap metrics for all three models, with full per-group performance metrics presented in Table C.2 in Appendix C.

Table 3.6: Fairness gap metrics by `sex_of_driver`.

Model	DP Difference	EO Difference
Logistic Regression	0.054	0.040
Random Forest	0.125	0.118
XGBoost	0.136	0.124

Logistic Regression exhibited the smallest sex-based disparity (DPD = 0.054, EOD = 0.040), while XGBoost showed the largest (DPD = 0.136, EOD = 0.124). Across all models, male drivers were assigned higher selection rates and higher recall than female drivers, suggesting that high-risk female drivers are more likely to be missed. This pattern is consistent with known gender imbalances in road accident datasets, where male drivers are overrepresented among serious collision records.

Fairness by Age Band

Fairness was then evaluated with `age_band_of_driver` as the sensitive attribute, where band 4 corresponds to drivers aged 17–20 and band 5 to drivers aged 21–25, both within the young driver population studied in this project. Table 3.7 summarises the results.

Table 3.7: Fairness gap metrics by `age_band_of_driver` (4 = 17–20, 5 = 21–25).

Model	Recall (Age 4)	Recall (Age 5)	DP Difference	EO Difference
Logistic Regression	0.690	0.515	0.147	0.175
Random Forest	0.706	0.540	0.133	0.166
XGBoost	0.601	0.452	0.089	0.149

Age-based disparities were notably larger than sex-based ones across all models. Younger drivers (band 4) consistently received higher recall than older drivers (band 5), meaning the models were more effective at identifying high-risk individuals within the target demographic. XGBoost showed the smallest age-based gap (DPD = 0.089), while Logistic Regression exhibited the largest (DPD = 0.147, EOD = 0.175).

Intersectional Fairness (Sex × Age)

To investigate whether fairness gaps get worse when combining both protected attributes together, an intersectional analysis was carried out. `sex_of_driver` and `age_band_of_driver` were combined into four subgroups: young male (1_age_4), older male (1_age_5), young female(2_age_4), and female older (2_age_5). Table 3.8 reports the resulting fairness gaps.

Table 3.8: Fairness gap metrics for intersectional groups (sex × age band).

Model	**DP Difference**	**EO Difference**
Logistic Regression	0.197	0.250
Random Forest	0.256	0.304
XGBoost	0.217	0.269

Intersectional gaps were larger than either sex or age gaps alone, showing that older female drivers experience the most unfair treatment — a pattern invisible when attributes are analysed separately. Random Forest showed the largest gaps (DPD = 0.256, EOD = 0.304), while Logistic Regression remained the fairest model across all three dimensions.

Bootstrap Confidence Intervals

To confirm the fairness metrics were stable and not just a result of the particular test set chosen, bootstrap resampling was performed using 200 resamples (`random_state=42`), producing 95% confidence intervals for DPD and EOD across all three fairness dimensions. Results are presented in Table 3.9.

Table 3.9: Bootstrap 95% confidence intervals for fairness metrics (200 resamples). DP = Demographic Parity Difference, EO = Equalized Odds Difference.

Model	**DP(sex)**	**95% CI**	**EO(sex)**	**95% CI**	**DP(age)**	**95% CI**
LR_balanced	0.053	[0.020, 0.095]	0.058	[0.014, 0.119]	0.146	[0.110, 0.175]
RandomForest	0.125	[0.092, 0.170]	0.131	[0.077, 0.210]	0.132	[0.100, 0.163]
XGBoost	0.137	[0.103, 0.183]	0.138	[0.085, 0.208]	0.087	[0.055, 0.123]

All fairness gaps are statistically stable across resamples. Logistic Regression's sex-based gap remains narrow (DPD = 0.053, 95% CI: [0.020, 0.095]), confirming it is consistently the fairest model by sex. Random Forest shows the largest intersectional instability (EOD = 0.304, 95% CI: [0.219, 0.453]), confirming the performance-fairness tradeoff is a consistent property of these models rather than a test-set artefact.

3.6 Streamlit Application Implementation

The demonstration application was implemented in `app.py` using Streamlit (v1.53.1), loading pre-trained models (`lr_balanced.pkl`, `rf_balanced.pkl`, `xgb.pkl`) and feature names at startup.

User Input and Prediction

Nine inputs are accepted: speed limit, road type, area type, lighting, weather, road surface, time of day, age of driver, and vehicle type. Sex and age band are collected separately in the sidebar exclusively for the fairness audit and are never passed to the model. Inputs are mapped to DfT STATS19 numeric codes and assembled into a 49-feature vector, with remaining features set to modal values derived from correctly predicted low-risk cases in the training set. The model outputs a probability score compared against the operational threshold ($\tau = 0.3$ for Random Forest and XGBoost, $\tau = 0.5$ for Logistic Regression) to generate a binary prediction.

SHAP Waterfall Explanation

Following each prediction, a SHAP waterfall plot is generated dynamically for the individual driver profile, showing how each feature pushed the prediction towards or away from high risk. The waterfall plot is rendered using `shap.plots.waterfall` with `st.pyplot`, providing a transparent, driver-specific explanation of every prediction made by the application.

Fairness Audit Display

A dedicated fairness audit section displays pre-computed DPD and EOD metrics for the selected model across sex and age band dimensions, using a traffic light indicator system:

GREEN = DPD/EOD < 0.05 (Fair) **AMBER** = 0.05–0.15 (Moderate) **RED** = > 0.15 (High disparity)

Plain English explanations accompany each metric, making fairness insights accessible to non-technical users.

3.7 Validation

Validation was conducted across four dimensions to ensure correctness, reproducibility, and reliability of the pipeline.

Data Pipeline Validation

All pipeline stages were validated through automated tests in `test_data_pipeline.py`, covering post-merge duplicate checks, age filter correctness, sex code cleaning, observation count, missing values, and protected attribute exclusion. The leakage validation table below confirms that performance drops after each removal were genuine — earlier configurations had been reading post-accident information rather than learning road safety patterns.

Table 3.10: ROC-AUC across pipeline versions confirming leakage removal.

Pipeline Version	Variables Removed	XGBoost ROC-AUC
Version 1 (initial)	None	> 0.95
Version 2	Severity columns	~ 0.75
Version 3 (final)	All leakage variables	0.689

Model and Fairness Validation

The stratified split preserved the 73%/27% class distribution in both partitions, confirmed by checking `value_counts(normalize=True)`. Fixed `random_state=42` produced identical results across multiple independent runs. Bootstrap resampling (200 iterations) confirmed all fairness gaps are statistically stable, as reported in Table 3.9.

Streamlit App Validation

The application was validated by confirming predictions matched notebook outputs for identical inputs, SHAP waterfall values summed correctly to the model output, and traffic light thresholds correctly reflected pre-computed fairness metrics.

Automated Tests

Five test files were implemented using `pytest`, each targeting a specific pipeline component. All tests pass successfully on Python 3.12.7, validating every stage of the pipeline.

Table 3.11: Summary of automated pipeline tests.

Test file	Coverage
`test_data_pipeline.py`	Validates age filter (17–25), sex code cleaning, binary target construction, absence of missing values, and protected attribute exclusion from feature matrix.
`test_leakage.py`	Confirms all severity-derived and identifier columns are absent from the feature matrix, ensuring no post-accident information leaks into model training.
`test_models.py`	Exercises all three models with a dummy input, verifying correct loading, probability output shape, values between 0 and 1, and binary prediction output.
`test_fairness.py`	Checks DPD and EOD fall within valid range on held-out test data and confirms a perfect fairness baseline produces DPD = 0 on controlled arrays.
`test_shap.py`	Verifies SHAP values have the correct shape and confirms the 3D dimensionality fix correctly extracts the high-risk class dimension.

User Evaluation

An informal usability evaluation was conducted with 3 non-technical participants, all young drivers aged 17–25 with active car insurance experience. Each participant was provided with a combined participant information and consent form before accessing the application. Participants interacted with the deployed Streamlit application independently via a shared URL before completing a structured 17-question questionnaire covering transparency, fairness, trust, and usability. Results are reported in Chapter 4, Section 4.5.

Chapter 4

Results, Evaluation and Discussion

4.1 Model Performance Results

Table 3.5 presents the final performance of all three models. Both ensemble models substantially outperformed Logistic Regression in terms of ROC-AUC (0.689 compared to 0.629), which shows that collision risk depends on non-linear feature interactions that linear models cannot capture. XGBoost had the highest accuracy (0.691) and precision (0.431), making it the most conservative model at flagging high-risk drivers. Random Forest had the highest recall (0.614), identifying the largest share of truly high-risk drivers, but also resulting in more false positives. These findings align with Masello et al. [4], who also report superior performance of gradient boosting methods for driver risk prediction. ROC curves for all models are shown in Figure 4.1.

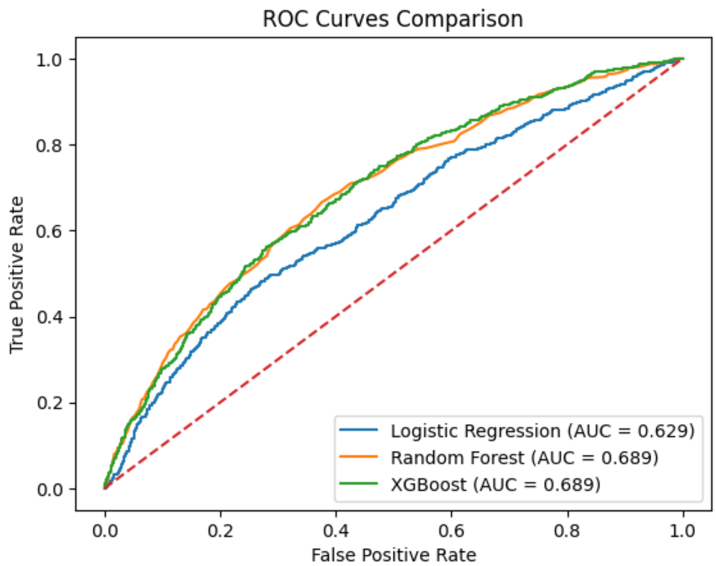

Figure 4.1: ROC curves for all three models on the held-out test set. Both ensemble models achieve ROC-AUC = 0.689, outperforming Logistic Regression (ROC-AUC = 0.629).

4.2 SHAP Explainability Results

A key finding is that `age_of_driver` ranks only 13th out of 49 features (mean |SHAP| = 0.032). Instead, road environment variables such as `police_force`, `number_of_vehicles`, `vehicle_type`, and `speed_limit` dominate predictions as shown in Figures 4.2 and 4.3. This suggests that contextual driving conditions contribute more to predicted risk than age alone, supporting concerns raised by Lindholm et al. [7] regarding potential indirect discrimination in age-based pricing.

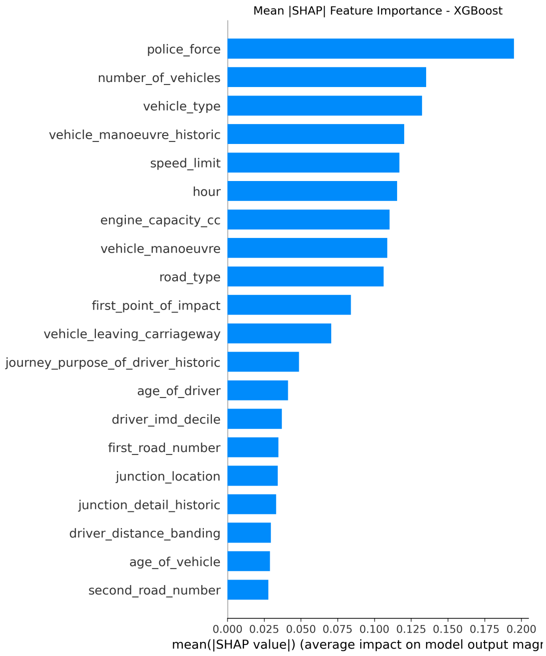

Figure 4.2: Global SHAP feature importance for XGBoost. `police_force` is the strongest predictor (mean |SHAP| = 0.195), while `age_of_driver` ranks 13th (mean |SHAP| = 0.032).

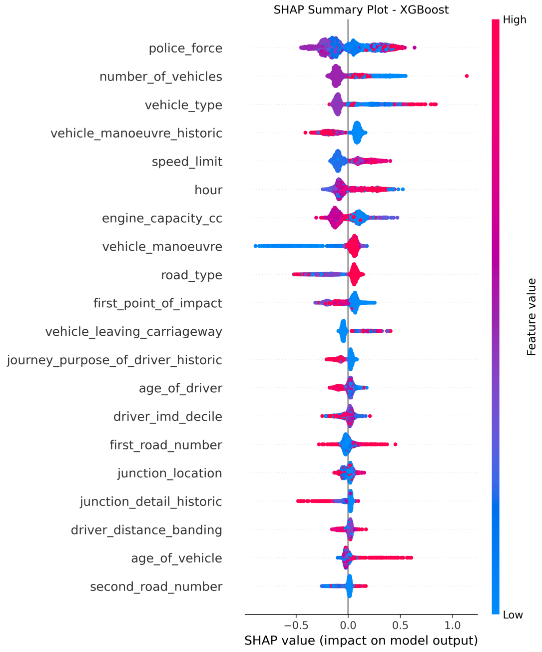

Figure 4.3: SHAP beeswarm plot showing direction and magnitude of each feature's effect. Red = high feature value, blue = low. Points spread right indicate positive impact on high-risk prediction.

4.3 Fairness Results and Discussion

The fairness evaluation revealed a consistent performance–fairness tradeoff across all three models, summarised in Tables 3.6, 3.7, and 3.8. No single model was fairest across both demographic dimensions simultaneously. Logistic Regression was the fairest by sex (DPD = 0.054) but least fair by age (DPD = 0.147), while XGBoost showed the opposite pattern. This directly confirms Hardt et al.'s [8] theoretical concern that optimising for predictive performance does not guarantee fairness across demographic groups.

Intersectional analysis reveals that fairness disparities compound when both attributes are considered together. Random Forest's intersectional DPD of 0.256 is more than double its sex-based DPD of 0.125, confirming that certain subgroups, particularly older female drivers, experience bias that is invisible when attributes are audited independently. This finding has direct implications for insurance pricing, where intersectional fairness auditing is rarely practised.

4.4 Evaluation Against Objectives and Deliverables

All six objectives and both deliverables were successfully met, as summarised in Table 4.1.

Table 4.1: Project objectives and deliverables evaluation.

ID	Objective / Deliverable	Status
O1	Dataset of 13,376 young driver observations constructed from DfT 2024 road safety data	Met
O2	Reproducible six-stage pipeline with Git version control and five automated test files	Met
O3	Three models trained and evaluated; XGBoost achieved highest accuracy (0.691) and ROC-AUC (0.689)	Met
O4	SHAP TreeExplainer applied to XGBoost, revealing `age_of_driver` ranks 13th out of 49 features	Met
O5	Fairness evaluated across sex, age band and their intersection using DPD and EOD with bootstrap validation	Met
O6	Interactive Streamlit application deployed with SHAP explanations and traffic light fairness audit	Met
D1	Project report documenting the full project lifecycle including research, methodology and evaluation	Met
D2	Software repository containing all technical artefacts, publicly available on GitHub	Met

4.5 User Evaluation

Three participants aged 17–25 completed the usability evaluation described in Section 3.7, interacting with the Streamlit application before completing a structured questionnaire. The mean Likert-scale responses are summarised in Table 4.2.

Table 4.2: Mean usability scores across three participants (1 = Strongly Disagree, 5 = Strongly Agree).

Statement	Mean
The risk prediction was easy to understand	4.67
The SHAP explanation helped me understand the prediction	4.33
I could identify which factors most influenced the prediction	4.67
The system appears to evaluate drivers fairly	2.67
The model does not rely unfairly on personal characteristics	3.00
The traffic light indicators were clear	5.00
I would trust the prediction	3.33
The interface was easy to use	5.00
I would find this tool useful for understanding my premium	4.67

The results indicate strong usability and interpretability. The interface and traffic light indicators received maximum scores (5.00), demonstrating that the system is accessible to non-expert users. High scores for prediction understanding (4.67) and factor identification (4.67) further confirm that SHAP explanations successfully support user interpretation.

In contrast, perceived fairness (2.67) and trust (3.33) were lower. This reflects a key finding: increased transparency does not necessarily increase user confidence. Instead, exposing fairness gaps prompted critical engagement with the system.

As one participant noted:

> "It changed my feeling from 'this is a cool tool' to 'this model needs more work before it can be used for real insurance pricing'."

This suggests the system achieves its intended goal of promoting informed scepticism rather than blind trust, an important requirement for responsible AI in high-stakes domains such as insurance. Participants also suggested improvements, including exposing more input features and extending explainability to all models. These indicate that while effective as a research tool, further refinement is needed for real-world deployment.

4.6 Limitations and Future Work

Three limitations are noted. The analysis is limited to drivers aged 17–25, preventing cross-cohort risk comparison. The dataset includes only road environment features, lacking behavioural data such as phone use or reaction time. The Streamlit demo uses 9 of 49 features, so predictions are illustrative rather than deployable.

The following steps would significantly enhance this framework:

Cross-cohort fairness analysis. Extending the pipeline to the full driver population would enable direct comparison of risk between young and older drivers, providing empirical evidence for or against age-based pricing. This would transform the framework from a proof-of-concept into a genuine policy evaluation tool.

Behaviour-based risk modelling. Using telematics and ADAS data, as demonstrated by Masello et al. [4], would enable individual driving behaviour, such as speed, braking, and phone use, to replace collision outcomes as the basis for risk prediction, producing fairer and more practical risk profiles.

Fairness-aware model training. Rather than auditing fairness post-hoc, future work could apply fairness constraints during training, or implement discrimination-free pricing formulas as proposed by Lindholm et al. [7], directly optimising for equitable outcomes across demographic groups.

Improved explainability interface. Expanding the Streamlit interface to show more features, use plain English labels, and add SHAP explanations for Logistic Regression would make the tool accessible and ready for real-world use.

4.7 Conclusion

This project demonstrated that transparent, fair machine learning can be applied to young driver motor insurance risk prediction using publicly available UK road safety data, with all six objectives met. Ensemble models outperformed Logistic Regression; SHAP analysis revealed `age_of_driver` ranked 13th of 49 features, challenging age-based pricing [7]; and no model achieved fairness across both sex and age simultaneously [8], with intersectional gaps larger than single-attribute analysis suggests.

FCA regulations require fair value in insurance pricing [13], and the EU AI Act classifies insurance risk assessment as high-risk, mandating transparency and fairness audits [14]. This project shows such audits are feasible with public data, representing a step towards the accountable, explainable AI that regulators and consumers increasingly demand.

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Appendix A

Self-appraisal

A.1 Critical self-evaluation

This project intended to create a transparent and fair machine learning framework for young driver motor insurance risk prediction, a topic that is personally relevant to me, since I passed my driving test in 2025, and right away experienced the high insurance premiums that young drivers face in the UK. All six objectives were met, and the framework developed shows that road environment features explain collision risk much better than age alone. This finding has real consequences for how insurers justify pricing based on age.

The biggest technical achievement was the improvement of the leakage detection process through several rounds of testing. Initially, the model exhibited unusually high ROC-AUC scores, above 0.95. This prompted a review of three versions of the pipeline. Post-accident variables, such as `number_of_casualties` and `did_police_officer_attend_scene`, were identified and removed, as they would not be available at prediction time. Careful consideration was given to the information that could actually be used for forecasting. Ultimately, this early challenge resulted in a final pipeline that was more robust and produced more reliable predictions.

The fairness evaluation produced more nuanced results than anticipated. No single model demonstrated fairness across both demographic dimensions simultaneously. Logistic Regression was fairest by sex but least fair by age, while XGBoost showed the opposite pattern. This tradeoff could not be resolved within the project; acknowledging this limitation is important. These findings demonstrate that deploying a model without a fairness audit can result in hidden biases in pricing decisions.

If the project were to be repeated, additional time would be allocated to the Streamlit application. The current demo utilises only 9 of 49 features, with the remaining 40 set to modal defaults, a limitation identified by all three usability participants. Expanding the interface to include additional features would enhance the usefulness of the transparency tools. Additionally, the deployment process was more complex than anticipated, with Python and library version compatibility issues consuming time that could have been spent on additional analysis.

A.2 Personal reflection and lessons learned

This project was quite a technically challenging piece of work undertaken during the degree. It became even more demanding because of the circumstances at the time. I was working on the project while also going through a placement interview process in December, which took up a lot of my time and energy. Managing both commitments simultaneously required careful prioritisation and reinforced the importance of time management and working under pressure, skills that are directly useful in a professional environment.

The most significant technical challenge was working without access to real insurance pricing data. Ideally, a project like this would use real premium records, claims histories, and underwriting variables from an actual insurer. This kind of data would let me directly test whether algorithmic pricing decisions are fair and justified. In reality, this data is commercially sensitive and not available to the public. So, I used the DfT STATS19 road safety dataset as the best available substitute. However, it only captures collision outcomes, not pricing decisions. This means the framework focuses on predicting risk instead of directly measuring premium fairness. This is an important difference, and future work with industry partners could help address it.

Even with this limitation, the project showed that road environment features explain collision risk much better than age alone. This finding is relevant to insurance pricing even without direct access to premium data. The absence of real insurance data also made the project more challenging in a productive way, requiring creative problem-solving

The most important lesson I learned was the value of working iteratively. The CRISP-DM framework I used in Chapter 2 was essential. Going back to earlier stages when I found leakage or when a model gave unexpected results was not a failure but a necessary part of rigorous ML development. This iterative mindset would be carried forward into any future data science work.

Working with real-world data rather than a clean benchmark dataset was also a significant learning experience. The DfT STATS19 dataset required substantial cleaning, merging, and leakage analysis before it was ready for modelling. This gave me a realistic idea of what production ML pipelines are like in practice.

On a personal level, this project reminded me how important it is to connect technical work to real-world impact. For example, finding that `age_of_driver` ranks 13th out of 49 features is not just a number. It is evidence that challenges a pricing practice that directly affects millions of young drivers in the UK. Putting the technical results in this context made the work feel genuinely meaningful instead of just academic.

A.3 Legal, social, ethical and professional issues

A.3.1 Legal Issues

This project used the UK Department for Transport STATS19 road safety dataset, published as open data under the Open Government Licence (OGL) v3.0 [15]. This licence permits free use, modification, and redistribution for any purpose, including academic research, provided the source is acknowledged. All use of this dataset complied fully with the OGL terms, and the source is cited throughout this report.

The user evaluation involved human participants and was therefore subject to the UK Data Protection Act (2018) and the General Data Protection Regulation (GDPR, 2018). The following measures ensured full compliance:

- No personally identifiable information was collected — responses were fully anonymous and no names were recorded.

- Participants were provided with a combined participant information and consent form prior to participation, clearly stating the purpose of the study, what data would be collected, and how it would be used.
- Data was collected solely for academic analysis and was not shared with any third party.
- No data was retained beyond what was necessary for the analysis, in accordance with the data minimisation principle under GDPR.

The Streamlit application does not collect, store, or transmit any user data. All inputs are processed within the session and discarded when the session ends, ensuring full compliance with GDPR data minimisation principles.

A.3.2 Social issues

The social implications of this project are significant. Algorithmic pricing in motor insurance is increasingly common in the UK, yet the models underlying these decisions are rarely transparent to consumers affected. Young drivers, who are among the most financially disadvantaged road users, face premiums that are often justified by age alone, without explanation.

This project demonstrates that road environment features better explain collision risk than age, directly challenging the social acceptability of age-based pricing as a primary rating factor. By making these findings accessible through an interactive demonstration, the project contributes to a broader conversation about insurers' social responsibility and consumers' rights to understand how algorithmic decisions are made about them.

The intersectional fairness analysis also revealed that older female drivers, a subgroup not typically highlighted in insurance fairness discussions, experience the largest model disparities when sex and age are considered together. This has social implications beyond the young-driver context, suggesting that fairness auditing should be standard practice for any insurance ML deployment.

A.3.3 Ethical issues

This project involved two categories of ethical consideration: the use of publicly available road safety data, and the conduct of a usability evaluation with human participants.

The usability evaluation falls under **Block Ethical Approval Category 3.4 (User Testing)**, as defined by the University of Leeds Faculty Research Ethics Committee. The following protocols were followed in accordance with this category:

- Each participant was provided with a combined participant information and consent form prior to participation, describing the purpose of the project, the data to be collected, and how it would be used.
- Participation was entirely voluntary. Participants were informed they could withdraw at any time before submitting their responses.
- Consent was implied by completion and submission of the questionnaire, in accordance with the University of Leeds guidelines for low-risk questionnaire-based research.

- All responses were anonymous. No names were collected and no participant is identifiable in this report. Three separate submissions can be confirmed via Google Forms timestamps.

A key ethical concern was the potential misuse of the tool to justify discriminatory pricing. This was mitigated by excluding protected attributes from the model, displaying a clear research-use notice in the application, and prominently highlighting model limitations in the fairness audit.
The dataset contains anonymised, publicly available records of serious accidents. Throughout, care was taken to respect the human context, with the aim of improving fairness and transparency in algorithmic decision-making.

A.3.4 Professional Issues

This project was conducted in accordance with the British Computer Society (BCS) Code of Conduct, which requires members to act in the public interest, maintain professional competence, and uphold the reputation of the profession. The following specific obligations were addressed:

Public interest The project prioritised transparency and fairness over predictive performance alone, reflecting the obligation to consider the broader societal impact of computing. The Streamlit application was designed to make responsible ML insights accessible to non-technical users, directly serving the public interest.

Professional competence All technical decisions were grounded in peer-reviewed literature, and all claims were supported by verified empirical evidence. Library versions were recorded in `requirements.txt` to ensure reproducibility, and automated tests were implemented using `pytest` to validate pipeline correctness.

Integrity Limitations of the framework were documented honestly throughout, including the performance–fairness tradeoff, the restricted feature set in the Streamlit demonstration, and the absence of real insurance pricing data. No results were overstated or presented without appropriate caveats.

Relevant authority The project engaged with existing regulatory frameworks including the FCA's principles for algorithmic fairness in financial services and the ICO's guidance on AI and data protection, ensuring the work is grounded in the current regulatory landscape for insurance ML in the UK.

Version control using Git with a feature branch strategy ensured that all implementation decisions were traceable and recoverable throughout development, reflecting professional software engineering practice.

Appendix B

External Material

The following external materials were used in this project. All other code, analysis, writing, and design is original work.

Datasets

DfT STATS19 Road Safety Data (2024) Published by the UK Department for Transport under the Open Government Licence v3.0. Two files were used: `dft-road-casualty-statistics- collision-2024.csv` and `dft-road-casualty-statistics- vehicle-2024.csv`. These were merged, filtered, and cleaned as described in Chapter 2. No modifications were made to the raw data beyond those documented in the pipeline. Available at: `https://www.data.gov.uk/dataset/road-accidents-safety-data`

Libraries and Frameworks

The following open-source libraries were used. All versions match those recorded in Table 3.1.

scikit-learn (v1.8.0) Used for Logistic Regression, Random Forest, StandardScaler, train-test split, and evaluation metrics.

XGBoost (v3.1.3) Used for gradient boosted classification.

SHAP (v0.50.0) [11]. Used for TreeExplainer and waterfall visualisations.

Fairlearn (v0.13.0) Used for MetricFrame, Demographic Parity Difference, and Equalized Odds Difference computation.

Streamlit (v1.53.1) Used for the interactive web application, deployed via Hugging Face Spaces.

pandas (v2.3.3), numpy (v2.3.5), matplotlib (v3.10.8), joblib (v1.5.3) Standard open-source Python libraries used for data manipulation, numerical computation, visualisation, and model serialisation respectively.

User Study Materials

The participant information and consent form used in the usability evaluation was adapted from the University of Leeds *Consent for Questionnaires and Surveys* template.[1] The questionnaire was designed by the author and administered via Google Forms.

[1] Available at: `https://secretariat.leeds.ac.uk/research-ethics/templates-and-documents-to-download/`

Deployment Platform

The Streamlit application was deployed using **Hugging Face Spaces**, a free hosting platform for machine learning demonstrations provided by Hugging Face Inc. The deployed application is publicly accessible at: `https://huggingface.co/spaces/sc23ss2/fyp-insurance-fairness` The underlying Docker-based Streamlit template provided by Hugging Face was used as the deployment environment.

Supervisor Contributions

Regular supervision meetings were held with Dr Arash Bozorgchenani throughout the project. One assessor meeting was held with Dr Eric Atwell. Feedback received during these meetings informed decisions on how to structure the literature review, the inclusion of additional diagrams to aid reader understanding, and the suggestion to produce a YouTube demonstration video.

Appendix C

Supplementary Materials

Threshold Evaluation Results

Table C.1: Full threshold evaluation results for Random Forest and XGBoost.

Model	Threshold	Accuracy	Recall	Precision	F1
Random Forest	0.5	0.741	0.104	0.544	0.175
Random Forest	0.4	0.735	0.336	0.499	0.401
Random Forest	**0.3**	**0.650**	**0.614**	**0.397**	**0.482**
Random Forest	0.2	0.463	0.875	0.315	0.463
XGBoost	0.5	0.744	0.131	0.571	0.213
XGBoost	0.4	0.730	0.291	0.485	0.363
XGBoost	**0.3**	**0.691**	**0.518**	**0.431**	**0.471**
XGBoost	0.2	0.534	0.820	0.342	0.482

Per-Group Fairness Performance

Table C.2: Per-group performance by `sex_of_driver` (1=male, 2=female).

Model	Group	Accuracy	Precision	Recall	F1	Selection Rate
LR_balanced	Male	0.572	0.360	0.600	0.450	0.487
	Female	0.593	0.262	0.566	0.359	0.433
RandomForest	Male	0.635	0.418	0.640	0.506	0.447
	Female	0.687	0.325	0.522	0.401	0.322
XGBoost	Male	0.668	0.444	0.545	0.490	0.358
	Female	0.746	0.381	0.421	0.400	0.222

Confusion Matrices

Confusion matrices for all three models at their selected decision thresholds are shown in Figures C.1, C.2, and C.3. Rows represent true labels (0 = low risk, 1 = high risk) and columns represent predicted labels.

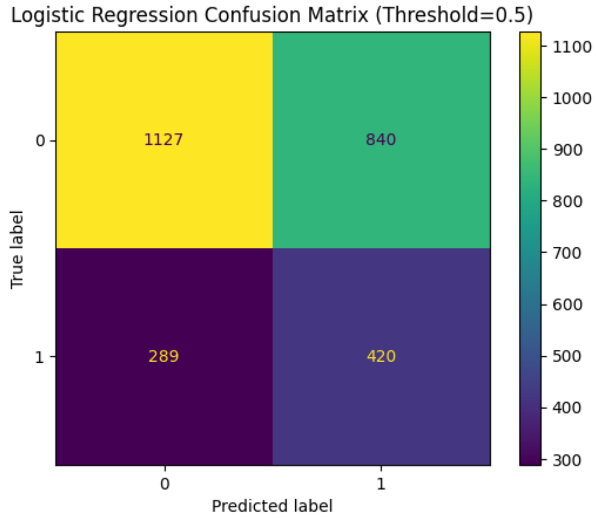


Figure C.1: Logistic Regression confusion matrix ($\tau = 0.5$).

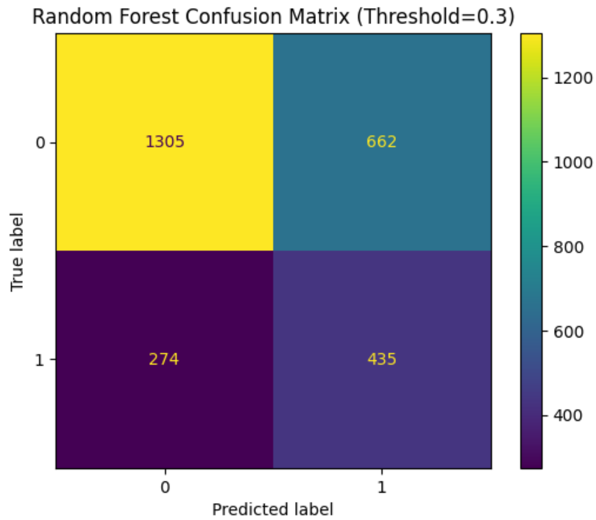


Figure C.2: Random Forest confusion matrix ($\tau = 0.3$).

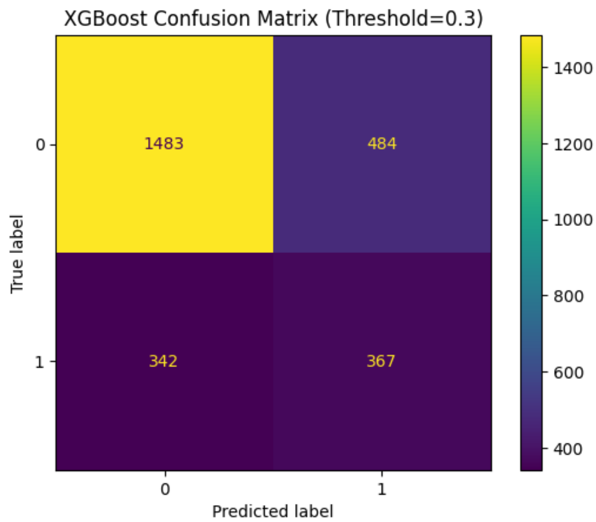


Figure C.3: XGBoost confusion matrix ($\tau = 0.3$).

Threshold Sweep Plots

Figures C.4, C.5, and C.6 show how precision, recall, and F1 vary across all evaluated thresholds for each model.

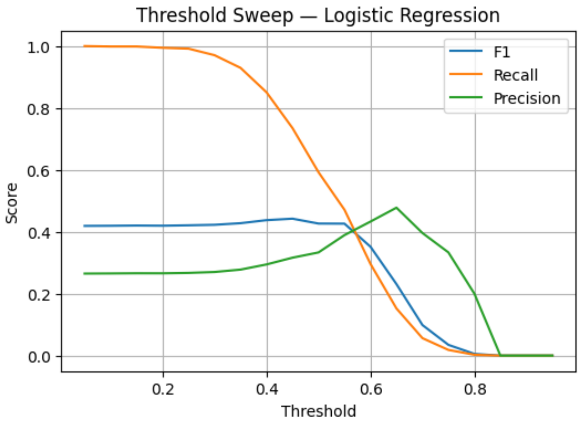


Figure C.4: Threshold sweep for Logistic Regression.

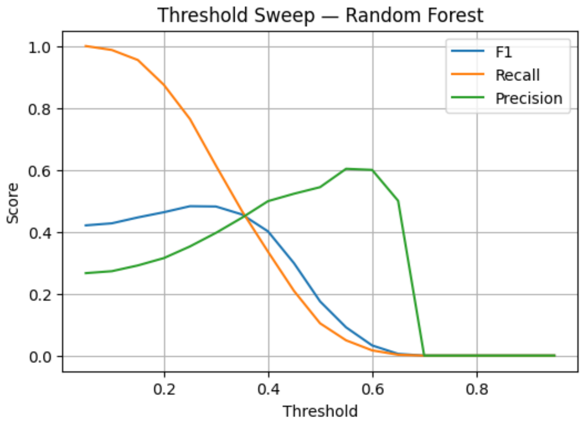


Figure C.5: Threshold sweep for Random Forest.

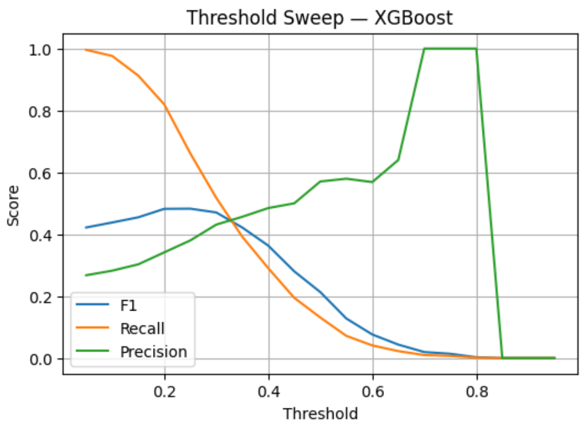


Figure C.6: Threshold sweep for XGBoost.

Repository Structure

Figure C.7 shows the final GitHub repository structure, with all pipeline notebooks, source code, tests, models, and data directories organised modularly as described in Section 2.1.

Project Structure

```
fyp-insurance-fairness-xai/
├── data/
│   └── processed/          # Cleaned datasets
│   └── raw/                # Uncleaned datasets
│   └── figures/            # Saved figures
├── models/                 # Trained model files (.pkl)
│   ├── feature_names.pkl
│   ├── lr_balanced.pkl
│   ├── rf_balanced.pkl
│   ├── xgb.pkl
├── notebooks/
│   ├── 01_data_loading_merge_audit.ipynb
│   ├── 02_feature_engineering_and_preprocessing.ipynb
│   ├── 03_baseline_model.ipynb
│   ├── 04_tree_baseline_models.ipynb
│   ├── 05_fairness_metrics.ipynb
│   └── 06_shap_explanations.ipynb
│   └── 07_save_models.ipynb
├── src/
│   └── app.py              # Streamlit application code
├── tests/
│   ├── __init__.py
│   ├── test_data_pipeline.py
│   ├── test_fairness.py
│   ├── test_leakage.py
│   ├── test_models.py
│   └── test_shap.py
├── README.md
└── requirements.txt
```

Figure C.7: Final GitHub repository structure showing modular organisation of notebooks, source code, tests, models, and data directories.

Final Kanban Board

Figure C.8 shows the completed GitHub Kanban board with all tasks successfully moved to the Done column, confirming project completion.

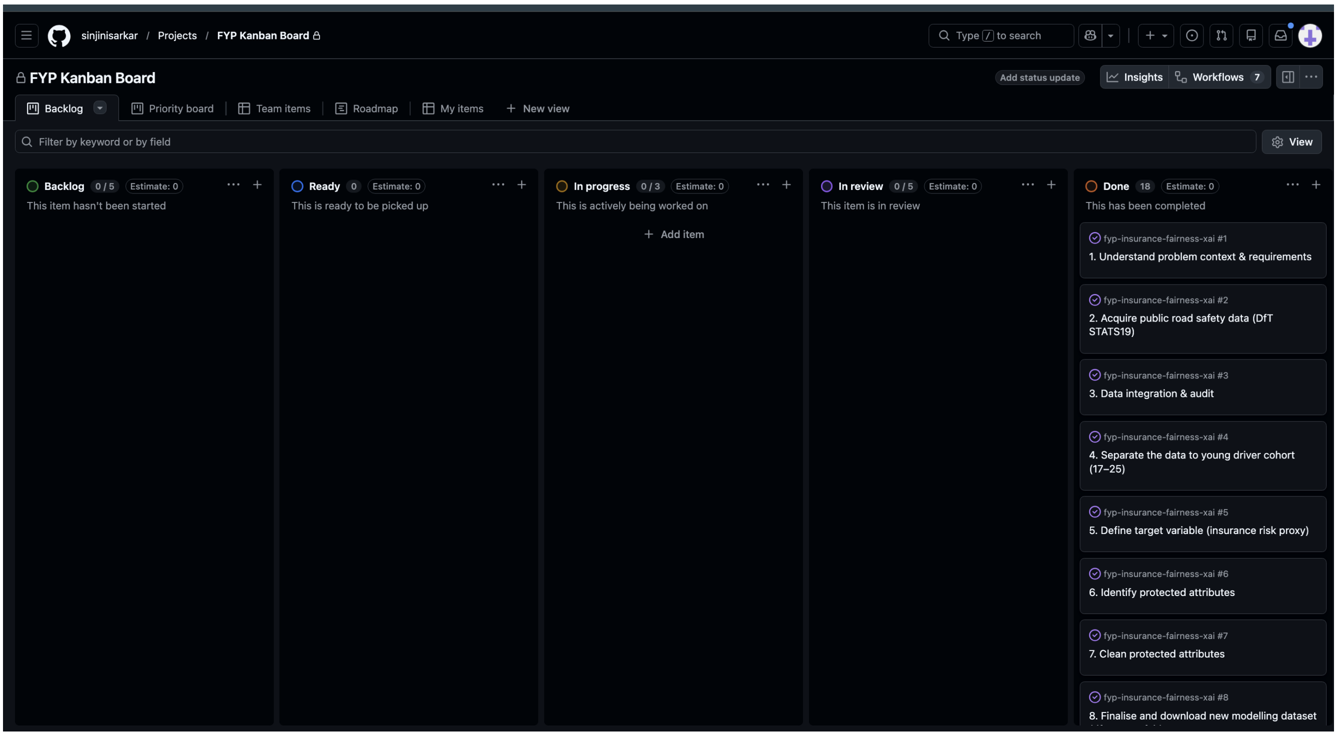


Figure C.8: Final GitHub Kanban board showing all tasks completed in the Done column.

User Study Materials

Participant Consent Form

The following consent form was provided to all three participants prior to completing the usability evaluation. Consent was implied by completion and submission of the questionnaire, in accordance with University of Leeds guidelines for low-risk research.

User Study — Participant Information & Consent Form

Final Year Project: Transparent and Fair Machine Learning for Young Driver Insurance Risk Prediction

School of Computing, University of Leeds | April 2026

Section 1: Participant Information

Thank you for taking part in this user study. Before you begin, please read the following information carefully.

Purpose of the study:

This study is part of a final-year undergraduate dissertation investigating the transparency and fairness of machine learning models used to predict motor insurance risk for young drivers in the UK. You will be asked to interact with an interactive web application (Streamlit) that shows how a model predicts insurance risk, explains its reasoning, and reports whether it treats different demographic groups fairly.

What you will be asked to do:

1. Open the Streamlit application using the link provided.
2. Enter some example driver and road condition details using the sidebar.
3. View the risk prediction, the SHAP explanation chart, and the fairness audit section.
4. Complete the short questionnaire below based on your experience.

Time required:

Approximately 10-15 minutes in total.

Confidentiality:

No personally identifiable information will be collected. All responses are anonymous and will only be used for academic purposes within this dissertation. Data will not be shared with any third parties.

Voluntary participation:

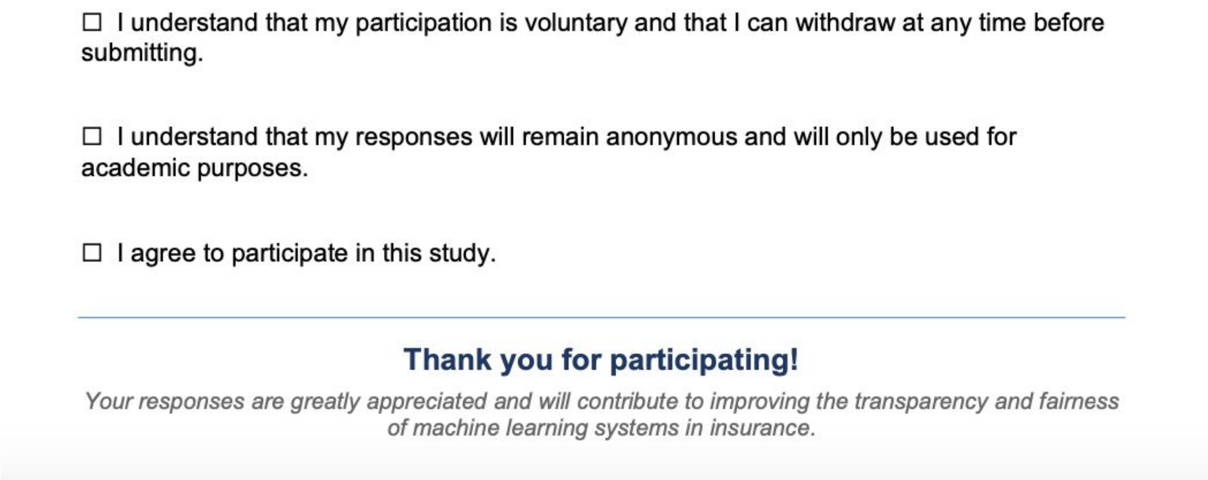
Participation is entirely voluntary. You may withdraw at any time before submitting your responses without giving a reason. You are not required to answer any questions you do not wish to.

Section 2: Consent

Please read each statement and tick the box to confirm your agreement before proceeding.

☐ I confirm that I have read and understood the participant information above.

Figure C.9: Participant information and consent form (page 1)



☐ I understand that my participation is voluntary and that I can withdraw at any time before submitting.

☐ I understand that my responses will remain anonymous and will only be used for academic purposes.

☐ I agree to participate in this study.

Thank you for participating!

Your responses are greatly appreciated and will contribute to improving the transparency and fairness of machine learning systems in insurance.

Figure C.10: Participant information and consent form (page 2)

Usability Evaluation Questionnaire

The questionnaire was administered anonymously via Google Forms. No questions were mandatory. The form is accessible at:

`https://forms.gle/JAFBpwvdPeeCGmE37`

The questions are reproduced below for reference. Background questions (1–4) used multiple choice. Questions 5–13 used a five-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree). Questions 14–17 were open-ended text responses.

Background

1. How old are you?
2. Do you currently own or drive a car?
3. Do you have car insurance?
4. Have you used ML/AI tools before?

Transparency and Explainability

1. The risk prediction provided by the system was easy to understand.
2. The SHAP explanation chart helped me understand why the prediction was made.
3. After viewing the explanation, I could identify which factors most influenced the prediction.

Fairness

1. The system appears to evaluate drivers fairly.
2. I feel the model does not rely unfairly on personal characteristics such as age or gender.

3. The traffic light fairness indicators (Green/Amber/Red) were clear and easy to understand.

Trust and Usability

1. I would trust the prediction provided by this system.
2. The interface was easy to use and navigate.
3. I would find this kind of tool useful for understanding my insurance premium.

Open Feedback

1. In your own words, what did the SHAP explanation tell you about why the prediction was made?
2. Did the fairness audit section change how you feel about the system? Please explain.
3. What did you like or dislike about the system?
4. Do you have any suggestions for how the system could be improved?

Anonymised Responses

Table C.3 summarises the Likert scale responses from all three participants.

Table C.3: Individual Likert scale responses from three usability participants (P1, P2 and P3) (1=Strongly Disagree, 5=Strongly Agree).

Statement	P1	P2	P3
Prediction easy to understand	5	4	5
SHAP explanation helpful	4	4	5
Could identify key factors	4	5	5
System appears fair	3	3	2
Model not unfairly reliant on personal characteristics	3	4	2
Traffic lights clear	5	5	5
Would trust the prediction	4	3	3
Interface easy to use	5	5	5
Would find tool useful	5	4	5

Open-Ended Responses

The following screenshots show the anonymised open-ended responses from all three participants, exported directly from Google Forms.

Open Feedback

14. In your own words, what did the SHAP explanation tell you about why the prediction was made?

3 responses

The SHAP chart showed that the number of vehicles involved was the biggest factor increasing my risk score, followed by the trunk road flag and police force region. Only the vehicle manoeuvre actually reduced my risk. It showed that road conditions matter more than personal characteristics like age.

The SHAP chart was very helpful because it broke down the risk into specific contributing factors. It showed me that the prediction wasn't just a random number; for example, I could see that the no. of vehicles involved and the police force area (region) were the biggest reasons my risk score was pushed higher. It also showed me factors that lowered my risk, like my Vehicle manoeuvre, which made the model feel much more transparent and less like a black box.

The SHAP explanation showed me which factors increased or decreased my risk score and by how much. The red and blue bars made it easy to see what was having a positive or negative impact on the prediction. It helped me understand exactly why I was considered higher or lower risk instead of just seeing a final result with no explanation.

Figure C.11: Question 14: What did the SHAP explanation tell you about why the prediction was made?

15. Did the fairness audit section change how you feel about the system? Please explain.

3 responses

Yes, it was surprising to see that age ranked only 13th out of 49 features, which challenges why young drivers are charged so much more. However the red flag for age group fairness was concerning, showing the model still treats 16-20 year olds differently to 21-25 year olds.

Yes, it definitely made me more cautious. While the prediction and the explanation were clear, the fairness audit revealed Moderate Disparity for both sex and age groups. Seeing that the model is more accurate for some age groups than others, and that it flags males as high risk more often, even when actual risk is similar, made me realise that the model still has inherent biases. It changed my feeling from 'this is a cool tool' to 'this model needs more work before it can be used for real insurance pricing'.

Yes, it did. At first the system felt fair because everything was explained clearly, but the fairness audit showed that there are still some differences in how groups are treated, especially with age. That made me realise that even if a model seems accurate, it can still have some bias. It made me trust the system more in a way, because it was honest about its limitations.

Figure C.12: Question 15: Did the fairness audit change how you feel about the system?

16. What did you like or dislike about the system?

3 responses

really liked the clean interface and how the steps were clearly numbered. The traffic light fairness indicators were very intuitive. I disliked that the SHAP chart wasn't available for Logistic Regression by default; it took switching models to see it.

I really liked how user-friendly the interface was, specifically the interactive sliders for age and time of day. The 'traffic light' system (Green/Amber/Red) in the fairness section made complex data very easy to understand at a glance. I disliked that 40 out of the 49 features used default values. While the SHAP chart was great, knowing so many factors were 'hidden' made me feel like I wasn't seeing the full picture of how my specific driving scenario might be judged.

I liked how clear and detailed the system was. It explained not only the prediction but also why it was made, which is something normal insurance websites don't do. I also liked the fairness section because it made me think about how these systems can affect different groups. One small downside is that some parts, like the fairness scores or technical terms, could be a bit confusing at first.

Figure C.13: Question 16: What did you like or dislike about the system?

17. Do you have any suggestions for how the system could be improved?

3 responses

It would be helpful if the SHAP chart was available for all models including Logistic Regression. Also a location/region input in the main form would be useful.

The system could be improved by allowing users to interact with more than just the current sidebar inputs, since the app mentions there are 49 features in total. Giving users more control over those 'default' values would make the transparency tools like SHAP even more useful. Also, adding a 'compare' feature to test the app's claim that road conditions explain risk better than age would be a great addition.

It could be improved by simplifying some of the technical info, especially the fairness scores, so it is much easier for non-technical users like me to understand. Adding short examples could help explain what those numbers mean inside the graph. It might also help to reduce the amount of information shown at once or guide the user step by step to make it feel less overwhelming. Overall it is a very detailed system and I really found it interesting to learn so much about insurance systems.

Figure C.14: Question 17: Suggestions for improvement.